

SUBHADEEP DUARI

PhD Candidate · Computational & Experimental Biology
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PROFILE

Computational biologist and hands-on experimentalist working at the interface of AI and the bench. I pair wet-lab molecular biology, protein biochemistry, and functional / biophysical assays with machine-learning models — producing analysis-ready experimental data and feeding it directly back into computational analysis to drive the next round of experiments. My published work spans AI-driven aging biology, deep-learning models of protein function, and ion-channel biophysics. Ph.D. in Computational Biology expected January 2027.

EDUCATION

Ph.D., Computational Biology — IIIT-Delhi 2022 – 2027 (expected)

Doctoral research at the computational–experimental interface · CSIR-UGC Research Fellow (JRF → SRF)

- Expected completion: January 2027.
- Research spans single-cell and nuclear morphometry, cellular aging — building interpretable ML models and validating predictions experimentally.

BS-MS, Biological Sciences — IISER Bhopal 2015 – 2020

- Five-year integrated Bachelor of Science–Master of Science; CGPA 7.05.

Senior Secondary (Class XII), Science — Midnapore Collegiate School, West Bengal 2013 – 2015

- Passed with 80.2%.

RESEARCH EXPERIENCE

Doctoral Researcher (Senior / Junior Research Fellow) 2022 – present

Department of Computational Biology, IIIT-Delhi

- Design, execute, and iterate on experimental and computational programs across single-cell imaging, nuclear morphometry, and cellular aging; build interpretable ML models and connect predictions back to bench validation.
- Generate analysis-ready imaging and phenotypic datasets and partner with computational biologists to iterate rapidly between experiment and analysis.
- Develop and apply deep-learning models for image processing and advanced image analysis; contribute to AI-for-biology platforms (aging prediction, generative molecular design) and manuscripts.
- Integrate LLM-based tools into experimental planning, protocol drafting, and data interpretation.

Project Research Assistant 2020 – 2021

Department of Biological Sciences, IISER Bhopal

- Characterized the activation mechanism of the TRPV1 ion channel by the spider-venom double-knot toxin (DkTx): recombinant protein production, membrane-affinity and bivalency characterization, and functional readouts by two-electrode voltage-clamp electrophysiology.

Summer Research Fellow 2016

Department of Mathematics, IISc Bangalore

- Worked in the field of category theory.

TECHNICAL SKILLS

Experimental biology: Molecular cloning; recombinant protein production; protein biochemistry and biophysical / functional characterization; two-electrode voltage-clamp (TEVC) electrophysiology; fluorescence microscopy; spectrofluorometry; mammalian cell culture; image-based phenotypic assays; HPLC; SDS-PAGE; DNA / RNA / plasmid isolation; agarose gel electrophoresis

Model systems: yeast, bacterial systems, *C. elegans*, and human cell lines.

Computational: Python, R; machine learning and deep learning; single-cell and image analysis; exploratory data analysis; familiarity with ML development practices

Databases & tools: Biological databases (UniProt, GenBank, PDB) and computational-biology tooling
AI-assisted research: LLM-based tools for experimental planning, protocol drafting, and data interpretation

PUBLICATIONS

Peer-reviewed articles

1. Duari S[†], Gautam V[†], Ahuja G. **Protocol for cellular age prediction in yeast and human single cells using transfer learning.** *STAR Protocols* (2025).
2. Gautam V[†], Duari S[†], Solanki S, Gupta M, Mittal A, Arora S, Aggarwal A, Sharma A K, Tyagi S, Pankajbhai R K. **scCamAge: A context-aware prediction engine for cellular age, aging-associated bioactivities, and morphometrics.** *Cell Reports* (2025).
3. Arora S[†], Mittal A[†], Duari S, Chauhan S, Dixit N K, Mohanty S K, Sharma A, Solanki S, Sharma A K, Gautam V. **Discovering geroprotectors through the explainable artificial intelligence-based platform AgeXtend.** *Nature Aging* (2025).
4. Mohanty S K[†], Mittal A[†], Farooqi N, Gaur A, Duari S, Solanki S, Sharma A K, Arora S, Kumar S, Gautam V., Dixit N K, Subramanian K, Ghosh T S, Sengupta D, Gupta S K, Murugan N A, Sharma D, Ahuja G. **Deep learning reveals endogenous sterols as allosteric modulators of the GPCR-Gα interface.** *eLife* (2025).
5. Kumar S[†], Solanki S[†], Gupta M[†], Chauhan S[†], Mohanty S K[†], Satija S, Sharma A, Duari S, Sharma A, Gautam V, Arora S, Shome R, Sinha S, Sharma A K, Mittal A, Sengupta D, Murugan N A, Ahuja G. **Chemical Dice Integrator (CDI): A scalable framework for multimodal molecular representation learning.** *Nature Communications* (2026)(in press).
6. Satija S[†], Mohanty S K[†], Jorvekar S[†], Verma A, Solanki S, Gautam V, Arora S, Mittal A, Sharma A, Chauhan S, Duari S, Kumar S, Rai A, Das A, Kakati K, Das K, Sarma A, Tayal J, Sengupta D, Mehta A, Borkar R M, Ahuja G. **MutAlverse: An AI-powered, mechanism-backed platform for discovering novel DNA adducts and their precursor genotoxins.** *Journal of Cheminformatics* (2026).
7. Singh Y, Sarkar D, Duari S, G S, Indra Guru P K, M V H, Singh D, Bhardwaj S, Kalia J. **Dissecting the contributions of membrane affinity and bivalency of the spider venom protein DkTx to its sustained mode of TRPV1 activation.** *Journal of Biological Chemistry* (2023)

Preprints & manuscripts

8. Solanki S[†], Mohanty S K[†], Satija S, Chauhan S, Bandaru N R, Dukare S, Tiwari N K, Kumar R N, Aravind A B, Mukherjee S, Chikkanna D, Balasubramanian W R, Sammeta S R, Gautam V, Arora S, Kumar S, Duari S, Sharma A, Shome R, Sengupta D, Abbineni C, Samajdar S, Ahuja G. Generative AI framework SynGlue for the rational design of clinically relevant protein degraders. *bioRxiv* (2025).
9. Ahuja G, Sharma A, Singh A, Kedia S, Sharma A, Gautam V, Sharma P, Khandelwal A, Ramakrishna S, Muddashetty R, Freude K, Solanki S, Chauhan S, Kumar S, Satija S, Duari S, Arora S, Gupta A, Shome R, Sengupta D. Cross-species aging knowledge integration into an agentic AI platform uncovers conserved mechanisms. *bioRxiv* (2026).
10. Chauhan S[†], Satija S[†], Solanki S[†], Duari S, Gautam V, Rakheja I, Kumar S, Arora S, Sharma A, Shome R, Sinha S, Ali S Y, Sengupta D, Murugan N A, Ahuja G. Liquid biopsy via calorimetric fingerprinting of plasma with inert probes for non-invasive cancer stratification. *Manuscript* (in preparation).

† Equal contribution.

CONFERENCE PRESENTATIONS

EMBO Immunology Sectoral Meeting 2025 — Marseille, France

29 – 30 Sep 2025

- Presented: “Extracting Biological Insights from Single-Cell Microscopic Images.”

EMBO Lecture Course: Post-transcriptional Regulation in Ageing and Age-related Diseases — Greater Noida & New Delhi, India

10 – 15 Jun 2024

- Presented: “Historical Mild Thermal Stress Pulsing (MTSP) Enhances Replicative Lifespan through Pro-longevity Metabolic Pathways.”

TEACHING EXPERIENCE

Teaching Assistant, IIIT-Delhi:

- Introduction to Quantitative Biology (IQB) — 2024
- Genetics and Molecular Biology (GMB) — 2024
- Data Science in Genomics (DSG) — 2023

AWARDS & SCHOLARSHIPS

CSIR-UGC Junior Research Fellowship — All India Rank 139	2019
DST-INSPIRE Scholarship — MHRD, Government of India	2015
Gold Medal, 21st National Child Science Congress — DST, Government of India	2014

LANGUAGES

English · Bengali · Hindi